Application of Euclidean Distance in K-Nearest Neighbor Regression

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An Example: Finding the K-Nearest Neighbors

Consider a two-dimensional feature space and the query point:

$$\mathbf{x}_q = (5, 5).$$

Suppose our training dataset contains the following 12 points:

Point	$\mathbf{x}_i$	y_i
$\mathbf{x}_1$	$(1, 2)$	10
$\mathbf{x}_2$	$(2, 8)$	16
$\mathbf{x}_3$	$(3, 4)$	12
$\mathbf{x}_4$	$(4, 6)$	14
$\mathbf{x}_5$	$(5, 3)$	11
$\mathbf{x}_6$	$(6, 5)$	15
$\mathbf{x}_7$	$(7, 7)$	17
$\mathbf{x}_8$	$(8, 4)$	18
$\mathbf{x}_9$	$(9, 6)$	20
$\mathbf{x}_{10}$	$(2, 3)$	9
$\mathbf{x}_{11}$	$(7, 3)$	16
$\mathbf{x}_{12}$	$(4, 3)$	13

We choose:

$$K = 3.$$

Step 1: Measure the Distance to Every Point

For the query $\mathbf{x}_q = (5, 5)$,

$$\mathbf{dist}(\mathbf{x}_q, \mathbf{x}_i) = \sqrt{(5 - x_i^{(1)})^2 + (5 - x_i^{(2)})^2}.$$

We calculate the distance to *every* one of the 12 training points:

Point	y_i	Distance	Point	y_i	Distance
$\mathbf{x}_1$	10	5.00	$\mathbf{x}_7$	17	2.83
$\mathbf{x}_2$	16	4.24	$\mathbf{x}_8$	18	3.16
$\mathbf{x}_3$	12	2.24	$\mathbf{x}_9$	20	4.12
$\mathbf{x}_4$	14	1.41	$\mathbf{x}_{10}$	9	3.61
$\mathbf{x}_5$	11	2.00	$\mathbf{x}_{11}$	16	2.83
$\mathbf{x}_6$	15	1.00	$\mathbf{x}_{12}$	13	2.24

Measure: Every training point is compared with the query.

Step 2: Select the K Nearest Neighbors

Since $K = 3$, select the three training points having the smallest Euclidean distances:

Neighbor	Target y_i	$\mathrm{dist}(\mathbf{x}_q, \mathbf{x}_i)$
$\mathbf{x}_6 = (6, 5)$	15	1.00
$\mathbf{x}_4 = (4, 6)$	14	1.41
$\mathbf{x}_5 = (5, 3)$	11	2.00

Therefore,

$$N_3(\mathbf{x}_q) = \{(\mathbf{x}_6, 15), (\mathbf{x}_4, 14), (\mathbf{x}_5, 11)\}.$$

Step 3: Predict

The KNN regression prediction is the arithmetic mean of the target values of the three nearest neighbors:

$$\hat{y}(\mathbf{x}_q) = \frac{15 + 14 + 11}{3} = \frac{40}{3} \approx 13.33.$$

Thus the KNN estimate for the unknown target at $\mathbf{x}_q = (5, 5)$ is:

$$\boxed{\hat{y}(\mathbf{x}_q) \approx 13.33}$$

The Essential Idea

The example illustrates the three stages of KNN regression:

$$\boxed{\begin{array}{c}\text{Measure}\\ \text{Euclidean distance}\end{array}} \longrightarrow \boxed{\begin{array}{c}\text{Select}\\ K \text{ nearest neighbors}\end{array}} \longrightarrow \boxed{\begin{array}{c}\text{Predict}\\ \text{average their target values}\end{array}}$$

Euclidean distance provides a geometric notion of proximity based on the inner product. KNN uses this geometry to construct a local estimate of an unknown function, based on the assumption that nearby points tend to have similar target values.

$$\boxed{\text{Inner Product}} \longrightarrow \boxed{\text{Norm}} \longrightarrow \boxed{\text{Distance}} \longrightarrow \boxed{\text{Nearest Neighbor}} \longrightarrow \boxed{\text{Prediction}}$$